UNIVERSIDAD DE LAS FUERZAS ARMADAS – ESPE
Department of Computer Science
Software Engineering
36862 – Object-Oriented Programming

UNIT 1 PROJECT · TEAM WORK

Project Definition: ProvStock

Prov&Tec Management, a Comprehensive Sales and Inventory System

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Revision Note

> *Version 1.2 updates version 1.1 so that it matches the Software Requirements Specification (SRS) v1.3, which records two decisions made during the implementation: the data is stored in JSON files, and the reports are exported to CSV files instead of PDF (Table 1 and Section 4). Version 1.1 had refined the problem statement, overview, and background and added the scope and key concepts sections; version 1.0 is kept unchanged as a record of the first iteration.*

1. Problem

Prov&Tec is an Ecuadorian company that sells electrical materials. It manages its sales and inventory manually, with paper documents (sales slips and order forms) and basic spreadsheets. This model has allowed the company to sustain itself and grow, but it no longer supports the daily control of the business and causes the following problems:

- **P1. Scattered information.** The data about customers, products, and suppliers is spread across papers and spreadsheets, with no single reliable source.
- **P2. Unknown real stock.** Incoming and outgoing goods are recorded late and inconsistently, and the location of the products in the warehouse is not recorded, so the staff do not know how many units of each product are available or where to find them.
- **P3. Unnecessary purchases.** Because the real stock is unknown, the company orders materials that it already has, which generates unnecessary expenses.
- **P4. Slow customer service.** Customers wait between 10 and 20 minutes while the salesperson searches for products, checks paperwork, and calculates the sale by hand, and many of them leave without buying.
- **P5. Uncontrolled restocking.** There is no timely warning of the products that are running out, which causes stockouts.
- **P6. Decisions without data.** Slow-moving stock accumulates and generates financial losses, and the owner has no sales history or reports to know what is sold, what is bought, and how much the inventory is worth.
- **P7. No traceability.** Several people with different responsibilities (salespeople, the warehouse clerk, and the owner) handle the same information without access control, and there is no record of who registered a sale, received goods, or corrected the stock.

Therefore, Prov&Tec needs a computer system that centralizes its information, keeps the stock up to date automatically, speeds up customer service, and gives the owner reliable data to control the business.

2. Overview

ProvStock is an object-oriented desktop application, developed in Java, that replaces the manual management of Prov&Tec. It centralizes the information about customers, products, suppliers, sales, and purchases; updates the stock of each product every time a sale or a purchase is registered; warns the staff when a product reaches its minimum stock level; and produces reports for the owner.

The system has three user roles. The **salesperson** serves customers at the counter, registers sales, and registers or updates customers. The **warehouse clerk** manages the products and their location in the warehouse, registers the goods received from suppliers, adjusts the stock, and receives the low-stock alerts. The **owner** can do everything the other two roles do and, in addition, manages user accounts and suppliers, voids sales, updates the VAT rate, reviews the sales history, and generates reports.

The system is organized in nine modules, one per Product Backlog item (Table 1).

Table 1
ProvStock Modules

Item	Module	What it does	Solves
RF01	Access Management	Login and logout, user roles and permissions, user accounts, password change and reset.	P7
RF02	Customer Management	Register, update, search, and deactivate customers; “Final Consumer” customer.	P1, P4
RF03	Product Management	Register, update, search, and deactivate products, with their minimum stock level and warehouse location.	P1, P2, P4
RF04	Sales Management	Register sales with automatic subtotal, VAT, and total; void sales; update the VAT rate.	P2, P4, P7
RF05	Inventory Management	Automatic stock updates, manual adjustments (including the initial inventory), and a record of every stock movement.	P2, P3, P7
RF06	Supplier and Purchase Management	Register, update, and deactivate suppliers; register purchases that increase the stock.	P1, P2, P7
RF07	Sales History	View and filter all sales, including voided ones, with their details.	P6, P7
RF08	Stock Alerts	Warn when a product is at or below its minimum stock level.	P3, P5
RF09	Reports	Sales, inventory, purchase, and slow-moving product reports; export to CSV.	P3, P6

3. Background

Since its beginning, Prov&Tec has operated with a manual management model. The company has managed to sustain itself, grow, and overcome significant obstacles, such as financial losses caused by stock that did not turn over, but its dependence on paper limits its efficiency. For this reason, the company wants to take a technological step to modernize its logistics, keep rigorous control of its inventory, and offer faster service.

In addition, the market is moving toward digital processes, and the administrative paperwork falls mainly on Ximena Añamise, who runs the business. A system that centralizes the information is the first step of that transition. Electronic invoicing with the Ecuadorian tax authority (SRI) is part of this trend and a possible future extension, but it is not included in this version of ProvStock.

As Object-Oriented Programming students at the University of the Armed Forces ESPE, the ByteWorks team chose this real problem to apply class and object modeling: the nouns of the problem became the objects and classes of the system (Section 5), and the requirements were specified in the SRS v1.3 following IEEE Std 830-1998.

4. Scope

Included in this version

- The nine modules of Table 1 (RF01 to RF09) and the three user roles.
- A desktop application in Java with an interface in Spanish, for Windows 10 or later, that works without an Internet connection.
- Data persistence in JSON files, so that the information is kept when the application is closed and reopened, and reports saved as CSV files that can be opened with Excel.

Outside the scope of this version

- Online sales and online payments.
- Electronic invoicing with the SRI and accounting.
- Management of several branches.
- A mobile application and barcode scanning.

Measurable goal: a trained salesperson registers a sale of up to five products in less than 2 minutes, compared with the current 10 to 20 minutes (SRS v1.3, NFR-03).

5. Key Concepts: From the Problem to the Classes

The classes of ProvStock were obtained from the problem: first, the nouns of the problem statement were identified; then, the nouns that represent things the system must store became objects, and the objects were grouped into classes. Table 2 shows that path. Class names are written in English, following the Java naming conventions.

Table 2
Nouns of the Problem and Resulting Classes

Noun in the problem	Class	Why
User, staff	`User`	Each person who uses the system needs an account (P7).
Role (salesperson, warehouse clerk, owner)	`Role`	Defines what each user can do (P7).
Customer, buyer	`Customer`	The person or company that buys (P1).
Supplier	`Supplier`	The company from which goods are bought (P1).
Product, electrical material	`Product`	What the company sells; it stores its stock and location (P2).
Sale, sales slip	`Sale`	Replaces the paper sales slip (P4).
Line of a sale	`SaleDetail`	Each product and quantity of a sale.
Purchase, order form	`Purchase`	Replaces the paper order form; increases the stock (P2).
Line of a purchase	`PurchaseDetail`	Each product, quantity, and cost of a purchase.
Incoming and outgoing goods	`StockMovement`	Records every change in the stock and who made it (P2, P7).
Reports	SalesReport, InventoryReport, PurchaseReport, SlowMovingProductsReport	Data for the owner’s decisions (P6).
Inventory	`Inventory`	Coordinates every stock change (P2).
Alert, stockout	`StockMonitor`	Detects the products at or below their minimum stock (P5).
Sales history	`SalesManager`	Keeps and filters the registered sales (P6).

Nouns that did not become classes

Some nouns became **attributes** of a class instead of classes: location, minimum stock, and price are attributes of Product (`location`, `minimumStock`, `salePrice`); payment method and VAT rate are attributes of Sale (`paymentMethod`, `vatRate`). Other nouns describe the current situation or the context and are **outside the system**: paper, spreadsheet, company, store,

warehouse, and market. The control classes AccessManager, CustomerManager, ProductCatalog, PurchasingManager, and ReportGenerator do not come from a noun of the problem; they were added in the design to coordinate each module, as explained in the UML Design Specification v1.3. The same applies to the classes added later for the files and to group data, such as FileManager, CsvReportExporter, ProductInfo, and SaleFilter.